

A causal model account of the Volitionality-Zero Result correlation¹

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Abstract. English lexical causatives typically entail result realization when used with a perfective aspect. A subset of these predicates, known as *defeasible causatives*, form an exception: with non-volitional subjects, they yield a *culminating* reading and *entail* the result, whereas with volitional subjects, they also allow a *zero result non-culminating* reading, in which the result is merely *implicated* and can be cancelled. This correlation between volitionality and the zero result non-culminating reading (*V-ZR correlation*) is captured by Demirdache and Martin's (2015) *Agent Control Hypothesis* (ACH). Most existing accounts adopt a *result cancellation approach*, assuming a single result-encoding predicate underlying both readings. Drawing on new data from Mandarin and English, I argue against this assumption and propose instead a *dual event structure approach*: the culminating reading involves an achievement predicate, which encodes a result, while the zero result non-culminating reading involves an activity or accomplishment predicate, which does not. This view, also advocated by Piñón (2014), leaves open why the zero result non-culminating reading typically requires a volitional subject. I address this question by developing a causal-model-based account.

Keywords: Agent Control Hypothesis, defeasible causatives, volitionality, non-culmination, event structure, causal model.

1. Introduction

Lexical causatives like *break* are commonly analyzed as bivalent, as in (1). They consist of a causing subevent (*e*) and a caused subevent (*e'*). The caused subevent represents the result or the culmination of the entire event described by the predicate.

$$(1) \quad \llbracket \text{break} \rrbracket = \lambda y \lambda e. \exists e'. [\text{cause}(e, e') \wedge \text{broken}(e') \wedge \text{theme}(e', y)]$$

When a lexical causative is used with a perfective aspect, the result is typically entailed.² Following Demirdache and Martin (2015), I call this the *culminating* reading. For example, when *break* appears in the simple past (commonly analyzed as the combination of the past tense and the perfective aspect), the associated result (the theme being broken) cannot be cancelled (see 2a). Thus, the predicate *break the window* entails the result that *the window is broken*. Entailment is indicated by an arrow in (2b).

¹ I would like to thank Magdalena Kaufmann, William Snyder, Stefan Kaufmann, and Jon Gajewski for their insightful comments on various versions of this paper. I also benefited greatly from feedback and discussion with the participants of the UConn Semantics Seminar, the UConn Ling Lunch, and SuB 30. I am grateful to Tyler Poisson, Robin Jenkins, William Snyder, and Walter Shaw for their judgments. Abbreviations used: CL = classifier; NEG = negation; PFV = perfective; PROG = progressive.

² Strictly speaking, in English, the perfective forms of *non-gradable* lexical causative verbs (e.g., *break*) generally entail the *complete* realization of the result, while the perfective forms of *gradable* lexical causative verbs (e.g., *burn*) generally entail at least the *partial* realization of the result (Demirdache and Martin 2015). When the realization is complete, the predicate receives a culminating reading. When the realization is partial, the predicate receives a partial result non-culminating reading.

- (2) a. Emma broke the window, #but it didn't break.
 b. Emma broke the window. → *The window was broken.*

However, it has been observed that certain lexical causatives, known as *defeasible causatives*, behave differently (Martin and Schäfer, 2012, 2013, 2017; Martin, 2015, 2020). *Teach* is one such verb. For these predicates, whether the result is entailed correlates with the volitionality of the subject, a pattern I refer to as the Volitionality-Zero Result Correlation (*V-ZR correlation*). When the subject is non-volitional, like *that textbook* in (4), the result is *entailed*, yielding a *culminating* reading.³ In contrast, when the subject is volitional, like *Sam* in (3), the result is merely *implicated* (indicated by a squiggly arrow), making available a *zero result non-culminating* reading in which the result can be fully cancelled.⁴

- (3) *Volitional subject, Zero result non-culminating reading possible*
 a. Sam taught me logic, but I didn't learn it at all.
 b. Sam taught me logic. ~→ *I knew logic.*
- (4) *Non-volitional subject, Zero result non-culminating reading impossible*
 a. That textbook taught me logic, #but I didn't learn it at all.
 b. That textbook taught me logic. → *I knew logic.*

Martin and Schäfer (2013) argue that defeasible causatives encode a sublexical modal operator. They contrast with *standard causatives* like *break*, which are extensional verbs that always entail their results in the perfective.

The V-ZR correlation has also been noted by Travis (2002), Bar-el (2005), Folli and Harley (2005), Jacobs (2011), Schäfer (2012), among others. It is formally captured by Demirdache and Martin's (2015) weak version of the Agent Control Hypothesis (ACH) in (5), with volitionality treated as a typical agenthood property.⁵

- (5) *Agent Control Hypothesis* (weak version)
 Zero result non-culminating construals ... require the predicate's external argument to be associated with 'agenthood' properties.
- (Demirdache and Martin 2015)

Two main approaches have been proposed to account for the V-ZR correlation: the *result cancellation approach* and the *dual event structure approach*. The result cancellation approach, adopted by most existing analyses (Martin and Schäfer, 2012, 2013, 2017; Martin, 2015, 2020;

³ What is entailed is that I knew at least some logic. The entailed result may be either *partial* or *complete* (see Demirdache and Martin 2015). For simplicity, this paper sets aside the partial result reading.

⁴ That double object verbs do not entail their encoded results when taking volitional subjects have been noticed by Oehrle (1976), Gropen et al. (1989), Baker (1997), Rappaport Hovav and Levin (2008), and Beavers (2011), among others.

⁵ The strong version of the ACH asserts that not only are zero result non-culminating readings possible only with agent subjects, but partial result non-culminating readings are also restricted to agent subjects. However, since the strong version is attested only in Salish languages and appears to involve a distinct phenomenon, I set this discussion aside.

Martin et al., 2021), assumes that a single predicate underlies both the culminating and the zero result non-culminating readings and that this predicate encodes a result. This assumption also underlies the ACH in (5). Under this approach, the central challenge is to explain why the encoded result can be fully cancelled only when the subject is volitional.

The dual event structure approach, by contrast, posits two predicates: the predicate under the culminating reading encodes a result, while the predicate under the zero result non-culminating reading does not (Travis, 2002; Folli and Harley, 2005; Schäfer, 2012; Piñón, 2014). The main question for this approach is why the predicate under the zero result non-culminating reading typically requires a volitional subject.

I argue that data from Mandarin and English favor the dual event structure approach. Eventuality type tests indicate that the predicate under the culminating reading is an achievement, while the predicate under the zero result non-culminating reading is an activity or an accomplishment. To explain the V-ZR correlation, I propose that the zero result predicate is related to the culminating predicate in a way that satisfies three conditions defined in a probabilistic causal model. This relationship is captured by a meaning postulate. The V-ZR correlation arises because volitional subjects can easily satisfy all three conditions, while non-volitional ones usually fail to satisfy one or more of them.

The rest of the paper is structured as follows. Section 2 reviews the two main approaches to the V-ZR correlation. Section 3 presents Mandarin and English data that support the dual event structure approach and argue against the result cancellation approach, showing that culminating readings involve achievements, whereas zero result non-culminating readings involve activities or accomplishments. Section 4 examines the relationship between the predicates under the two readings and argues that the derivational direction is irrelevant to the V-ZR correlation. Section 5 develops a causal-model-based account of the V-ZR correlation. Section 6 concludes.

2. Previous approaches to the V-ZR correlation

As noted in the introduction, two main approaches have been proposed to account for the V-ZR correlation: the *result cancellation approach* and the *dual event structure approach*. Section 2.1 presents a representative result cancellation analysis, namely sublexical modality analyses. Section 2.2 outlines the dual event structure approach.

2.1. Result cancellation approach

The result cancellation approach assumes that a single result-encoding predicate underlies both the culminating and the zero result non-culminating readings. Under this approach, the V-ZR correlation can be understood as reflecting volitionality's tendency to cancel the predicate's encoded result. Most (if not all) analyses in this tradition explain this correlation by positing a modal component that shifts the predicate's encoded result to non-actual worlds. Sublexical modality analyses locate this modal component at the lexical level.

These analyses treat defeasible causatives as containing a sublexical modal operator (Koenig and Davis 2001), which scopes over the caused subevent. Under these analyses, *teach* can be represented as in (6). Martin and Schäfer (2017: 101) propose that the modal operator $\Box_p$ quantifies over all *causally successful* worlds, namely “those worlds that have duplicates of the event described by the verb, where the encoded [result] is obtained, and where the ‘conditions of success’ that are possibly associated with the event described by the verb are fulfilled.” Since the actual world need not be included in the modal operator’s domain of quantification, the result does not have to obtain in reality.

- (6) $\llbracket \text{teach} \rrbracket = \lambda y \lambda z \lambda e. [\text{teach}(e) \wedge \text{recipient}(e, y) \wedge \text{theme}(e, z) \wedge \Box_p \exists e'. [\text{cause}(e, e') \wedge \text{know}(e') \wedge \text{holder}(e', y) \wedge \text{theme}(e', z)]]$
 (adapted from Martin 2020: 267)

However, modality alone is not sufficient to account for the V-ZR correlation. Martin (2020: 284) further proposes the following two pragmatic principles.

- (7) a. The event type denoted by defeasible causative VPs used agentively is tokenized by actions of the subject’s referent.
 b. The event type denoted by defeasible causative VPs used non-agentively is tokenized by changes-of-state of the theme’s referent.

(7) indicates that agentivity affects how we decide whether an event belongs to the denotation of a predicate. Take the verb *teach* as an example. When it takes a volitional subject (an agent), the subject’s actions directed towards achieving the goal that the theme learns something qualify as tokens of the event type $\lambda e...[\text{teach}(e)...]$, by virtue of (7a). Since these actions do not entail the result that the theme learns something, and the sublexical modal operator allows the encoded result to fail to obtain in the actual world, a zero result non-culminating reading is possible. In contrast, when *teach* takes a non-volitional subject (a non-agent), teaching subevents must involve a change-of-state in the theme from not knowing something to knowing it, by virtue of (7b). Although the sublexical modal operator still allows the result not to occur, the teaching subevents already entail at least part of it, making the zero result non-culminating reading unavailable.⁶

2.2. Dual event structure approach

The dual event structure approach treats defeasible causatives as ambiguous between two lexical entries. Crucially, the one under the culminating reading and the one under the zero result non-culminating reading represent different event structures or eventuality types (see Martin and Schäfer 2013 for a discussion).

⁶ Sublexical modality analyses are not the only analyses within the result cancellation approach. Weak perfective aspect analyses also fall under this category. While sublexical modality analyses locate the modal component at the lexical level, weak perfective aspect analyses place it at the clause level, attributing modality to the perfective aspect. These analyses have been proposed to explain the V-ZR correlation in languages such as Mandarin, Hindi, and Thai, which, unlike English, also permit zero result non-culminating readings for standard causatives. Due to space limitations, I refer the reader to Martin (2020) for an overview.

Oehrle's (1976, section 4.2) analysis of *teach* belongs to this approach: one lexical entry of this verb "corresponds to an activity associated with a goal," which does not entail the result associated with the verb, while the other "is roughly equivalent to CAUSE Y to know/learn," which entails the verb's result. Only the activity use requires the subject to be volitional.

Piñón (2014) proposes that defeasible causatives are activities or accomplishments when they take agents (volitional subjects) and are achievements (or states) when they take causers (non-volitional subjects). While activities do not entail a result and thus allow zero result non-culminating readings, accomplishments and achievements entail a result, yielding culminating readings. However, no analysis has yet explained why the event structure under the zero result non-culminating reading typically requires a volitional subject.

The present work provides further evidence for the dual event structure approach, while adopting a different view on the eventuality types of the predicates under the two readings. I argue that the predicate under the culminating reading is an achievement and can take either volitional or non-volitional subjects, while the predicate under the zero result non-culminating reading is an activity or an accomplishment and requires a volitional subject.

3. Problems for the result cancellation approach

Data from Mandarin and English challenge the result cancellation approach and instead support the dual event structure approach, showing that the predicates under the zero result non-culminating and the culminating readings exhibit distinct aspectual properties. Section 3.1 and 3.2 discuss the Mandarin and English data, respectively. Section 3.3 presents eventuality tests from Mandarin that further distinguish the predicates under the two readings.

3.1. Mandarin bare and resultative forms

The pattern observed in (3–4) for the English defeasible causative *teach* is also attested in Mandarin. However, in Mandarin the V-ZR correlation arises across two morphologically related but aspectually distinct verbal forms. Consider the Mandarin counterpart of *teach*. The bare verb *jiao* is compatible only with volitional subjects, such as *Sam* in (8a), but does not allow non-volitional subjects, such as 'that book' in (8b). Notably, when the resultative morpheme *hui* 'know' is added, both volitional and non-volitional subjects are possible, as shown in (8c–d).

- (8) a. Sam jiao-le wo luojixue, dan wo genben mei xue-hui.
 Sam teach-PFV me logic but I at.all NEG study-know
 'Sam taught me logic, but I didn't learn it at all.'
- b. #na ben shu jiao-le wo luojixue.
 that CL book teach-PFV me logic
 Intended: 'That book taught me logic.'
- c. Sam jiao-**hui**-le wo luojixue, #dan wo genben mei xue-hui.
 Sam teach-**know**-PFV me logic but I at.all NEG study-know
 'Sam made me learn logic by teaching it to me, #but I didn't learn it at all.'

- d. na ben shu jiao-**hui**-le wo luojixue, #danwo genben mei xue-hui.
 that CL book teach-**know**-PFV me logic but I at.all NEG study-know
 ‘That book made me learn logic, #but I didn’t learn it at all.’

The implication pattern is given in (9), which closely parallels that shown by *teach* in (3–4), reproduced in (10). If *teach* with a volitional subject is assumed to be ambiguous, corresponding either to the bare form *jiao* ‘teach’ or the resultative form *jiao-hui* ‘teach-know’ in Mandarin, the implicature in (10a) can be understood as arising from the interaction of the non-entailment in (9a) and the entailment in (9b). With a non-volitional subject, *teach* corresponds only to the resultative form, yielding the entailment in (10b), as also seen in (9c).

(9) *Mandarin*

- a. “Sam taught me logic.” →/ “*I knew logic.*”
 b. “Sam taught-know me logic.” → “*I knew logic.*”
 c. “That book taught-know me logic.” → “*I knew logic.*”

(10) *English*

- a. Sam taught me logic. ↗ *I knew logic.*
 b. That textbook taught me logic. → *I knew logic.*

Mandarin bare and resultative forms are not only morphologically distinct but also show different aspectual properties with respect to before-clauses and the adverb *chadian* ‘almost.’ *Before*-clauses locate the reference time at some point of the event described by the predicate they contain. When the predicate contains a bare form like ‘teach,’ the *before*-clause picks out the event’s beginning point (see 11a). In contrast, when the predicate contains a resultative form like ‘teach-know,’ the *before*-clause picks out the event’s end point regardless of the subject’s volitionality, and crucially, cannot target the beginning point (see 11b–c).

(11) *Before-clauses*

- a. zaiSam jiao wo luojixue zhiqian, ...
 at Sam teach me logic before
 ✓beginning point reading: before Sam started to teach me logic, ...
 ✗end point reading: before Sam made me fully learn logic, ...
 b. zaiSam jiao-**hui** wo luojixue zhiqian, ...
 at Sam teach-**know**me logic zhiqian
 ✗beginning point reading: before Sam started to teach me logic, ...
 ✓end point reading: before Sam made me fully learn logic, ...
 c. zaina ben shu jiao-**hui** wo luojixue zhiqian, ...
 at that CL book teach-**know**me logic before
 ✗beginning point reading: before that book started to teach me logic, ...
 ✓end point reading: before that book made me fully learn logic, ...

The adverb ‘almost’ gives patterns similar to *before*-clauses. When modifying a bare verb like ‘teach,’ as in (12a), ‘almost’ gives rise to a *counterfactual* reading, denying that the event occurred at all. In contrast, when modifying a resultative form like ‘teach-know,’ as in (12b–c), ‘almost’ yields an *incomplete* reading, implying that the event started but did not culminate in the intended result.

(12) Adverb ‘almost’

- a. Sam chadian jiao-le wo luojixue.
 Sam almost teach-PFV me logic
 ✓ *counterfactual reading*: Sam didn’t even start to teach me logic.
 ✗ *incompletive reading*: Sam failed to make me fully learn logic.
- b. Sam chadian jiao-**hui**-le wo luojixue.
 Sam almost teach-**know**-PFV me logic
 ✗ *counterfactual reading*: Sam didn’t even start to teach me logic.
 ✓ *incompletive reading*: Sam failed to make me fully learn logic.
- c. na ben shu chadian jiao-**hui**-le wo luojixue.
 that CL book almost teach-**know**-PFV me logic
 ✗ *counterfactual reading*: That book didn’t even start to teach me logic.
 ✓ *incompletive reading*: That book failed to make me fully learn logic.

Data like (8) further support the V-ZR correlation while challenging the result cancellation approach, since they clearly involve two aspectually distinct predicates, undermining the single predicate assumption central to the result cancellation approach. Additional verb pairs that behave like ‘teach’ and ‘teach-know’ include *anwei* ‘comfort’ vs. *anwei-dao* ‘comfort-arrive,’ *xia* ‘scare’ vs. *xia-dao* ‘scare-arrive,’ *bangzhu* ‘help’ vs. *bangzhu-dao* ‘help-arrive,’ *cang* ‘hide’ vs. *cang-qilai* ‘hide-arise,’ *xiu* ‘fix’ vs. *xiu-hao* ‘fix-good.’

3.2. English defeasible causatives

Applying the *before*-clause and *almost* tests to English defeasible causatives reveals that their aspectual properties vary with the subject’s volitionality, further challenging the result cancellation approach.

If, as suggested in the previous section, English *teach* corresponds to Mandarin bare and resultative forms when taking a volitional subject, but only to the resultative form when taking a non-volitional subject, we predict that *teach* should allow both readings with *before*-clauses and with *almost* when the subject is volitional; with a non-volitional subject, only the reading associated with the resultative form should be available.⁷ This expectation is indeed met, as shown in (13) and (14).

(13) *Before*-clauses

- a. before Sam taught me logic, ...
 ✓ *beginning point reading*: before Sam started to teach me logic, ...
 ✓ *end point reading*: before Sam made me fully learn logic, ...
- b. before that textbook taught me logic, ...
 ✗ *beginning point reading*: before that textbook started to teach me logic, ...
 ✓ *end point reading*: before that textbook made me fully learn logic, ...

⁷ The observation that counterfactual readings of *almost* available with lexical causatives when they take volitional subjects disappear when the same verbs take non-volitional subjects has been noted by Vecchiato (2004) and Martin (2015).

(14) Adverb *almost*

- a. Sam almost taught me logic.

✓*counterfactual reading*: Sam didn't even start to teach me logic.

✓*incompletive reading*: Sam failed to make me fully learn logic.

- b. That textbook almost taught me logic.

✗*counterfactual reading*: That textbook didn't even start to teach me logic.

✓*incompletive reading*: That textbook failed to make me fully learn logic.

3.3. Aspectual classification of bare and resultative forms

Section 3.1 showed that Mandarin bare and resultative forms have distinct aspectual properties. In this section, I apply eventuality type tests to show that bare forms are activities or accomplishments, while resultative forms are achievements.

Following Rothstein (2004), I classify eventuality types into states, activities, accomplishments, and achievements based on two features: [$\pm$ telic] and [$\pm$ stages]. (15) illustrates this classification.

- (15) a. *states*: [$-$ telic, $-$ stages]
 b. *activities*: [$-$ telic, $+$ stages]
 c. *accomplishments*: [$+$ telic, $+$ stages]
 d. *achievements*: [$+$ telic, $-$ stages]

I use *for X time* adverbials and *in X time* adverbials to test for telicity. Atelic predicates are compatible with the former but not the latter, while telic predicates are compatible with the latter but not the former. The bare verb *jiao* 'teach' can be either telic or atelic (16a–b), while the resultative form *jiao-hui* 'teach-know' is telic (16c–d).

- (16) a. Sam zai yi zhou nei jiao-le wo luojixue.
 Sam at one week in teach-PFV me logic
 'Sam taught me logic in a week.'
 b. Sam jiao-le wo yi xiaoshi luojixue.
 Sam teach-PFV me one hour logic
 'Sam taught me logic for an hour.'
 c. Sam zai yi zhou nei jiao-**hui**-le wo luojixue.
 Sam at one week in teach-**know**-PFV me lgic
 'Sam made me learn logic in a week.'
 d. *Samjiao-**hui**-le wo yi xiaoshi luojixue.
 Sam teach-**know**-PFV me one hour logic

Rothstein (2004) links the feature [$\pm$ stages] to a predicate's ability to appear with the progressive. As shown in (17), the bare verb 'teach' can appear with the progressive, but the resultative form 'teach-know' cannot, indicating that the former is [$+$ stages] and the latter is [$-$ stages].

- (17) a. Sam zheng zai jiao wo luojixue.
 Sam just PROG teach me logic
 ‘Sam is teaching me logic.’
 b. *Sam zheng zai jiao-**hui** wo luojixue.
 Sam just PROG teach-**know**me logic

Piñón (1997) argues that achievements are incompatible with manner adverbs. I suggest that only [+stages] predicates can take manner adverbs. As shown below, the bare form ‘teach’ can cooccur with manner adverbs such as ‘seriously’ (18a), while its resultative correlate cannot (18b). Therefore, manner adverbs yield the same pattern as the progressive with respect to the [±stages] feature: bare forms are [+stages], while resultative forms are [–stages].

- (18) a. Sam renzhende jiao-le wo luojixue.
 Sam seriously teach-PFV me logic
 ‘Sam taught me logic seriously.’
 b. *Sam renzhende jiao-**hui**-le wo luojixue.
 Sam seriously teach-**know**-PFV me logic

The result is summarized as follows. Accordingly, *jiao* ‘teach’ is an activity or an accomplishment, and *jiao-hui* ‘teach-know’ is an achievement.

- (19) a. *jiao* ‘teach’: [±telic, +stages]
 b. *jiao-hui* ‘teach-know’: [+telic, –stages]

4. Relationship between the two event structures

Data from Mandarin and English suggest that two predicates of distinct eventuality types are involved, supporting the dual event structure approach to the V-ZR correlation. This raises the question of how the two event structures are related: they may be independent, or one may derive from the other. The latter option seems more economical. Mandarin data suggest that the predicate under the zero result non-culminating reading is more basic, with the predicate under the culminating reading derived via the addition of a resultative. However, as shown below, there are also cases in which the apparent direction of derivation is reversed.⁸ I argue that the derivational direction is irrelevant to explaining the V-ZR correlation.

As discussed in the introduction, standard causatives like *break* do not show the V-ZR correlation when used with a perfective aspect. However, as observed by Martin (2015), standard causatives display a similar pattern to the Mandarin data in (8) when comparing their progressive and simple past uses. When used with the progressive, *break* is more ready to take a volitional subject like *Emma* in (20a). A non-volitional subject like *the storm* in (20b) is only possible if the breaking is imminent (the *imminent* reading) or if cracks are visibly spreading on the window at the reference time (the *partial result* reading). Crucially, (20b) is infelicitous in a context where the storm was blowing at the window and the window broke a few minutes

⁸ The relationship between the predicates under the two readings is reminiscent of that observed in causativization, where the causative may be derived from the inchoative (*causativization*), the inchoative from the causative (*anti-causativization*), or both may share the same form (*labile*). For relevant discussion see Haspelmath (2016).

later. When the predicate is used with the progressive, as in (20a), the result is cancellable. When the predicate is used with the simple past, as in (20c) and (20d), both volitional and non-volitional subjects are acceptable, and the result is entailed.⁹

- (20) a. Emma was breaking the window, but it didn't break.
 b. (#)The storm was breaking the window.
 c. Emma broke the window, #but it didn't break.
 d. The storm broke the window, #but it didn't break.
- (21) a. Emma was breaking the window. →/ *The window was broken.*
 b. Emma broke the window. → *The window was broken.*
 c. The storm broke the window. → *The window was broken.*

If the progressive requires the predicate it takes to be [+stages], achievements like *break* must first be coerced into activities or accomplishments to satisfy the felicity conditions of the progressive. This suggests that the predicate in the progressive, which has the zero result non-culminating reading, is derived from the predicate in the perfective, which has the culminating reading. Reversing the derivational direction would make it difficult to explain why a predicate belonging to an aspectual class compatible with the progressive must be coerced into a different class when appearing in the simple past. This would be surprising as the simple past does not generally impose restrictions on the aspectual class of the predicates with which it combines.

The Mandarin data in (22) support the same conclusion. The verb *zhe* 'snap' is an achievement by default. It is telic, as indicated by its compatibility with the adverbial 'in an instant,' and it allows both volitional and non-volitional subjects, as shown in (22c) and (22d). However, when *zhe* 'snap' is coerced into an activity by the adverbial 'for a long time' in (22a) and (22b), only volitional subjects are possible. The implication pattern parallels that of progressive achievements in English; cf. (20) and (21). This further suggests that the predicate under the culminating reading is basic, while the predicate under the zero result non-culminating reading is derived.

- (22) a. wo zhe-le na gen shuzhi hao jiu, dan mei duan.
 I snap-PFV that CL branch very long but NEG break
 'I tried to snap that branch for a long time, but it didn't break.'
- b. #da.feng zhe-le na gen shuzhi hao jiu.
 big.wind snap-PFV that CL branch very long
 Literally: 'The big wind snapped that branch for a long time.'
- c. wo yizhayandegongfu jiu zhe-le na gen shuzhi,
 I in.an.instant at.once snap-PFV that CL branch
 #dan mei duan.
 but NEG break
 'I snapped that branch in an instant, #but it didn't break.'

⁹ Martin (2015) observes that when the subject is volitional, standard causatives in the progressive allow zero result non-culminating readings; when the subject is non-volitional, the result must be at least partially realized (the partial result reading). However, she does not discuss the imminent reading or the general infelicity of sentences like (20b).

- d. da.feng yizhayandegongfu jiu zhe-le na gen shuzhi,
big.wind in.an.instant at.once snap-PFV that CL branch
#dan mei duan.
but NEG break
‘The big wind snapped that branch in an instant, #but it didn’t break.’

(23) *Mandarin*

- a. “I snapped that branch for a long time.” →/ “*The branch was broken.*”
b. “I snapped that branch in an instant.” → “*The branch was broken.*”
c. “The big wind snapped that branch in an instant.” → “*The branch was broken.*”

Since there is no correlation between morphological markedness and the predicates under the two readings, I conclude that the derivational direction is irrelevant to explaining the V-ZR correlation.¹⁰

5. A probabilistic causal model account

As argued in section 3, data from Mandarin and English support the dual event structure approach, under which the predicate associated with the culminating reading encodes a result, while the predicate associated with the zero result non-culminating reading does not. Explaining the V-ZR correlation therefore reduces to explaining why the latter predicate requires a volitional subject, while the former allows both volitional and non-volitional subjects. I develop such an account using a probabilistic causal model in this section.

Section 5.1 introduces causal models and presents the core proposal. Section 5.2 applies the account to the data discussed earlier. Section 5.3 addresses apparent exceptions where the predicate under the zero result non-culminating reading does not require a volitional subject.

5.1. Causal models and the proposed account

Causal models, originally developed in statistics and given their influential formalization in Pearl (2000/2009), have in recent years found fruitful applications in linguistics (Schulz, 2011; Kaufmann, 2013; Lassiter, 2017; Mizuno and Yang, 2018; Alonso-Ovalle and Hsieh, 2021; Nadathur, 2023; Nadathur and Bar-Asher Siegal, 2024, among others). Following Lassiter (2017) and Mizuno and Yang (2018), I adopt a *probabilistic causal model* and specify it as follows.

(24) A probabilistic causal model M is a quadruple $\langle S, E, D, \mu \rangle$, where:

- a. S is a set of situations. b. E is a set of events.
c. D is a causal structure (i.e., a directed acyclic graph) $\langle V, \Rightarrow \rangle$, where V is a set of variables (or equivalently, a set of finite partitions on S) and $\Rightarrow$ encodes direct causal relations among them.

¹⁰ There are two additional patterns not discussed here, both of which belong to the *equipollent* category. The pairs may involve *suppletion* (like *treat/cure* in English), or both members include a marker (like control transitivizers and non-control transitivizers in Salish languages, see, e.g., Bar-el, 2005; Jacobs, 2011).

- d. μ is a probability measure mapping each proposition (obtained by assigning a value to a variable) to a value in the interval $[0,1]$.
- (25) a. The parents of variable X are given by $pa(X) = \{Y \mid Y \Rightarrow X\}$.
 b. The descendants of variable X are given by $de(X) = \{Y \mid X \Rightarrow^T Y \text{ or } Y = X\}$ where the relation $\Rightarrow^T$ is the transitive closure of $\Rightarrow$.¹¹

I argue that an achievement predicate P is related to its activity or accomplishment counterpart P' (roughly P -in-progress) in the model by a meaning postulate, with the direction of *morphological* derivation being irrelevant, as discussed in section 4. P places minimal constraints on its subject, allowing both volitional and non-volitional subjects. In contrast, P' must satisfy three causal-model-based conditions which specify the relationship between P' and P . These conditions are more easily satisfied when the event involves a volitional participant, but not when it involves a non-volitional one. Therefore, P' requires its subject to be volitional. Specifically, I propose that $P'(e)(s)$ is true only if the following three conditions are met, where e is an event and s is a situation:

- (26) a. *Causal relevance condition:*
 The situation s verifies a variable causally relevant to the result encoded by P .
 b. *Probabilistic bias condition:*
 Given the situation s , the result encoded by P is likely but not guaranteed to occur in the actual world.
 c. *Participant identity condition:*
 The event e involves the same participants as at least one P -event in a possible situation.

This is formalized in (27). Let Q and Q_i be variables. Recall that a variable is a finite partition on S . Following Kaufmann (2013), I assume for simplicity that all partitions on S are bipartitions. Such partitions can be defined as functions from $\{0, 1\}$ to propositions (i.e., sets of situations). For example, a variable Q_i can be represented as $\{q_i, \neg q_i\}$, where q_i is the proposition that answers Q_i affirmatively, and $\neg q_i$ the proposition that answers it negatively. Thus $Q_i(1) = q_i$ and $Q_i(0) = \neg q_i$. The set A_X of causal ancestors of $X \in V$ is given by $A_X = \{Y \in V \mid Y \Rightarrow^T X\}$. θ is a function that returns the ordered set of participants of an event.

- (27) Let e be an event, s a situation, P an achievement predicate, P' a P -in-progress predicate, C a variable encoding the result of P , and define $c = \lambda s. \exists e. [P_C(e)(s)]$, where P_C denotes the result state predicate encoded by P . Then:

$$P'(e)(s) \rightarrow \exists Q. [s \in Q(1) \wedge Q \in A_C]$$

$$\wedge 0.5 < \mu(c \mid \{p \mid p \in Q_i \wedge Q_i \in A_C \wedge s \in p\}) < 1$$

$$\wedge \forall e' \forall s'. [P'(e')(s') \rightarrow \exists e'' \exists s''. [P(e'')(s'') \wedge \theta(e') = \theta(e'')]]$$

(27) states that an event e is a P' -event in an situation s only if (i) the positive answer to a causal ancestor of C is true in the situation s ; (ii) given the propositions that are causally relevant to

¹¹ The transitive closure of a binary relation R on a set S is the smallest transitive relation R^T on S such that $R \subseteq R^T$.

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C in s , the probability of c is greater than chance (>0.5) but less than certainty (<1); and (iii) e involves the same ordered participants as at least one P -event in some possible situation s ".

Consider the achievement *Sam-teach-know-me-logic* (P) and the activity or accomplishment correlate *Sam-teach-me-logic* (P'). Assume a toy causal structure with $V = \{W, X, K\}$, where W stands for “whether Sam wants me to know logic,” X for “whether Sam explains logic to me,” and K for “whether I know logic.” P encodes the result k (i.e., the proposition “I know logic”). The direct causal relation $\Rightarrow$ denotes the set $\{\langle W, X \rangle, \langle X, K \rangle\}$. The causal structure can be represented as follows:

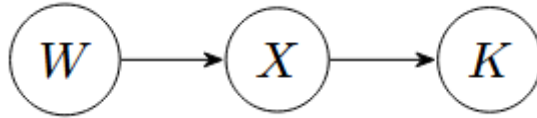


Figure 1: The causal structure underlying *Sam-teach-know-me-logic*.

In a situation where the proposition w (“Sam wants me to know logic”) and the proposition x (“Sam explains logic to me”) are true, two causal ancestors of the culmination variable K are verified (i.e., $s \in W(1)$, $s \in X(1)$). Therefore, the causal relevance condition is satisfied. Assume the prior probability distribution of the relevant propositions in the model is $\mu(w) = \mu(x) = 0.2$, $\mu(k | w, x) = 0.99$, $\mu(k | \neg w, x) = 0.5$, $\mu(k | w, \neg x) = 0.05$, $\mu(k | \neg w, \neg x) = 0.01$. Given w and x , the probability of k is high but not certain (i.e., 0.99), satisfying the probabilistic bias condition. P -events involve the ordered participants $\langle \text{Sam}, \text{me}, \text{logic} \rangle$. By the participant identity condition, any event in this situation that shares the ordered participants is a P' -event.

This proposal is reminiscent of Martin’s (2015) account of the V-ZR correlation, which builds on Varasdi’s (2014) analysis of the progressive. In brief, Martin proposes that the zero result non-culminating reading arises only when an event is “more than a try, less than a success” of the predicate. The probabilistic bias condition corresponds to Martin’s (2015) revision of Varasdi’s (2014) *indicative condition*, which states that an event e indicates P if and only if e realizes significantly more *sine qua non* (Sqn) conditions for e to culminate in P than not. In causal terms, Sqn conditions can be seen as propositions true in a situation that increase the likelihood of culmination. The probabilistic bias condition also subsumes Varasdi’s (2014) *sustaining condition*, which requires e to continue towards the culmination. In other words, given the situation, the probability of culmination is greater than 0.

5.2. Deriving the V-ZR correlation

While achievements are compatible with both volitional and non-volitional subjects, their activity or accomplishment counterparts typically occur only with volitional subjects. I propose that this restriction arises because non-volitional subjects generally fail to satisfy either the probabilistic bias condition or the participant identity condition, except in certain limited circumstances to be discussed in section 5.3.

5.2.1. Violation of the probabilistic bias condition

Consider first cases where the probabilistic bias condition is violated. I argue that the volition or intention of an agent typically ensures that the probability of culmination exceeds 0.5, while its absence tends to lower the probability below 0.5. I suggest that there are two reasons for this.

First, causal networks representing predicates with volitional subjects are generally more accessible to ordinary observers. Human actions and intentions make it much easier to predict how an event will unfold than when its development depends on forces or physical mechanisms involving inanimate objects. In other words, when a volitional subject is involved, we can readily construct a causal network that specifies how the result is achieved and which variables are relevant. In contrast, with a non-volitional subject, it may be unclear which variables matter at all for bringing about the result.

Second, observers generally have richer and more detailed information about humans than about inanimate entities. Since the probability of culmination depends on how much information we have about the variables (namely their values in a situation) in the causal network, the more certain and extensive our knowledge of the causal ancestors of the culmination, the easier it becomes to predict the probability of culmination, which is the prerequisite for satisfying the probabilistic bias condition.

Let us now examine how violation of the probabilistic bias condition explains the data discussed earlier. Recall that the Mandarin bare verb ‘teach’ can only take volitional subjects (28). Consider events that satisfy the participant identity condition, namely events involving the same ordered participants as those described by the predicate ‘teach-know me logic.’ This excludes events like me studying logic. When the subject is volitional, as in (28a), we have clear knowledge of the types of actions relevant to bringing about the intended result. Sam might explain concepts, write examples on the board, or test my understanding, and if one approach fails, he can adjust his teaching strategy (rephrase, elaborate, or use new materials) until I understand. Such goal-directed behavior increases the likelihood that I will learn logic. Consequently, the probability of culmination approaches 1, making it felicitous to say (28a), even if I ultimately did not learn it. In contrast, when the subject is non-volitional, as in (28b), there is no intentional process linking the cause and the result. The mere presence of an open book in front of me does not reliably predict whether I will learn its contents. As a result, the probability of culmination remains near or below chance, making it infelicitous to describe the situation using the bare verb ‘teach.’

- (28) *Mandarin*
a. “Sam taught me logic.”
b. “#That book taught me logic.”

The pattern in (29) can be explained similarly. When a volitional argument is involved, we generally know what kinds of activities are relevant to breaking a window (hammering it, hitting it with a stick, or throwing rocks at it, among many other possibilities). If the initial attempt fails, the agent can change their method (exert more force or use a more suitable tool). As a result, the probability that the window would break is close to 1, rendering it felicitous to

say (29a). In contrast, when a non-volitional argument is involved, it is often unclear which variables are relevant to the breaking of a window (wind speed, temperature, or air pressure). Even when such variables can be identified, ordinary observers are typically uncertain about their exact values. Therefore, when one observes a storm blowing at a window, the probability that the window will break is lower than chance, making it unwarranted to say (29b). However, as noted in section 4, (29b) becomes felicitous if the speaker knows that the window was about to break or that the result was partially realized (e.g., visible cracks suggesting partial breakage) at the reference time. In both cases, the probabilistic bias condition remains satisfied.

- (29) a. Emma was breaking the window.
b. (#)The storm was breaking the window.

5.2.2. Violation of the participant identity condition

Let us turn to cases where the participant identity condition is violated. The motivation behind this condition is that the argument structure of the original predicate should be preserved: the derived predicate inherits the arguments from the base predicate, and these arguments should retain their original θ -roles. This presupposes a particular theory of θ -roles, which lies beyond the scope of the present paper. For current purposes, it is sufficient to assume that the participants of the derived predicate must match those of the base predicate and appear in the same hierarchical order (roughly: initiator¹² > affectee > theme > source).

Consider (28) again. In a situation where I was studying logic from a book, the probability that I ended up knowing logic is high, satisfying the probabilistic bias condition. Nonetheless, it is still infelicitous to say (28b). The reason is that the event of my studying logic from a book involves an initiator ‘I,’ a theme ‘logic,’ and a source ‘that book,’ while the event of the book making me learn logic involves a different ordered set: an initiator ‘that book,’ an affectee ‘I,’ and a theme ‘logic.’ In other words, the participant identity condition is violated.

- (30) $\theta(e) = \langle I, \text{logic}, \text{that book} \rangle \neq \theta(e') = \langle \text{that book}, I, \text{logic} \rangle$

5.3. Explaining special cases

So far, cases illustrating the V-ZR correlation have contrasted volitional and non-volitional subjects, with volitionality treated as an agenthood property. However, agentivity does not always entail volitionality. One exception involves *instrument subjects*, namely objects that act only under the control of a volitional agent. Planes are instruments in this sense, since they require pilots to operate them. As shown in (31), the progressive form of the achievement *land* is acceptable with the instrument subject *the plane*. Although the progressive imposes an agentivity restriction on the subject, this restriction does not entail volitionality in the case of instrument subjects.

¹² Initiator subsumes both agent and causer (see Ramchand 2008).

- (31) *Instruments as agents*
 a. The plane landed.
 b. The plane was landing.

Another exception involves *machine subjects*, which are non-volitional entities that act according to pre-set programs. (32) shows that while the progressive form of *wake up Ana* is felicitous with volitional subjects like *I* (32a), it is infelicitous with non-volitional subjects like *the dishwasher's noise* (32b) unless under the imminent reading or the partial result reading. Machine subjects like *the alarm clock*, however, pattern with volitional subjects (32c).

- (32) *Machines as agents*
 a. I'm waking up Ana.
 b. (#)The dishwasher's noise is waking up Ana.
 c. The alarm clock is waking up Ana.¹³

I argue that progressive uses of achievements can take non-volitional subjects when these are instruments or machines, because, in these cases, the probabilistic bias condition is fulfilled: the updated probability of the encoded result is relatively high. For instruments, although they lack volition themselves, they presuppose a volitional manipulator. From the presence and behavior of an instrument, observers can infer the intentions or plans of this implicit agent, which strongly predict the event's outcome. In (31b), the plane is operated by a pilot. Observing the plane's descent, it can be inferred that the pilot has initiated and intends to complete the landing, making the probability of culmination nearly certain. In contrast, if the pilot is incapacitated, the plane no longer functions as an instrument, and its descent does not reliably predict a safe landing, rendering (31b) infelicitous in this scenario.

Machines require a different explanation. Although they can act autonomously, their behavior reflects prior human design aimed at reliably realizing a specific function. For example, alarm clocks are designed to wake sleepers. Observing an alarm clock sounding makes waking highly probable, accounting for the felicity of (32c). A dishwasher's noise, in contrast, is not designed to wake people and provides no reliable basis for predicting that outcome, causing the probabilistic bias condition to fail and making (32b) generally infelicitous.

6. Conclusion

The Volitionality-Zero Result (V-ZR) correlation captures the generalization that predicates under the zero result non-culminating reading typically require a volitional subject. Two main approaches have been proposed to account for this correlation: the result cancellation approach and the dual event structure approach. Most analyses in the literature adopts the result cancellation approach, but new data from Mandarin and English support the dual event structure approach. To explain the V-ZR correlation, I argue that the predicate under the zero result non-culminating reading (*P'*) is related to the predicate under the culminating reading (*P*) in the model by a meaning postulate. *P'(e)(s)* is true only if three conditions defined in a probabilistic causal model are fulfilled. First, the situation *s* verifies a variable causally relevant to the result encoded by *P* (*causal relevance condition*). Second, given the situation *s*, the result

¹³ These examples are inspired by Martin (2015).

encoded by *P* is likely but not guaranteed to occur in the actual world (*probabilistic bias condition*). Third, the event *e* involves the same participants as at least one *P*-event in a possible situation (*participant identity condition*). The V-ZR correlation arises because volitional subjects can easily satisfy all three conditions, while non-volitional ones usually fail to satisfy either the second or the third condition. This work sheds new light on the nature of agentivity. Focusing on three canonical agentive arguments (volitional agents, instruments, and machines), it shows that they all share the property of making the development of the event they engage in reliably predictable. This provides a new way of viewing agentivity.

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